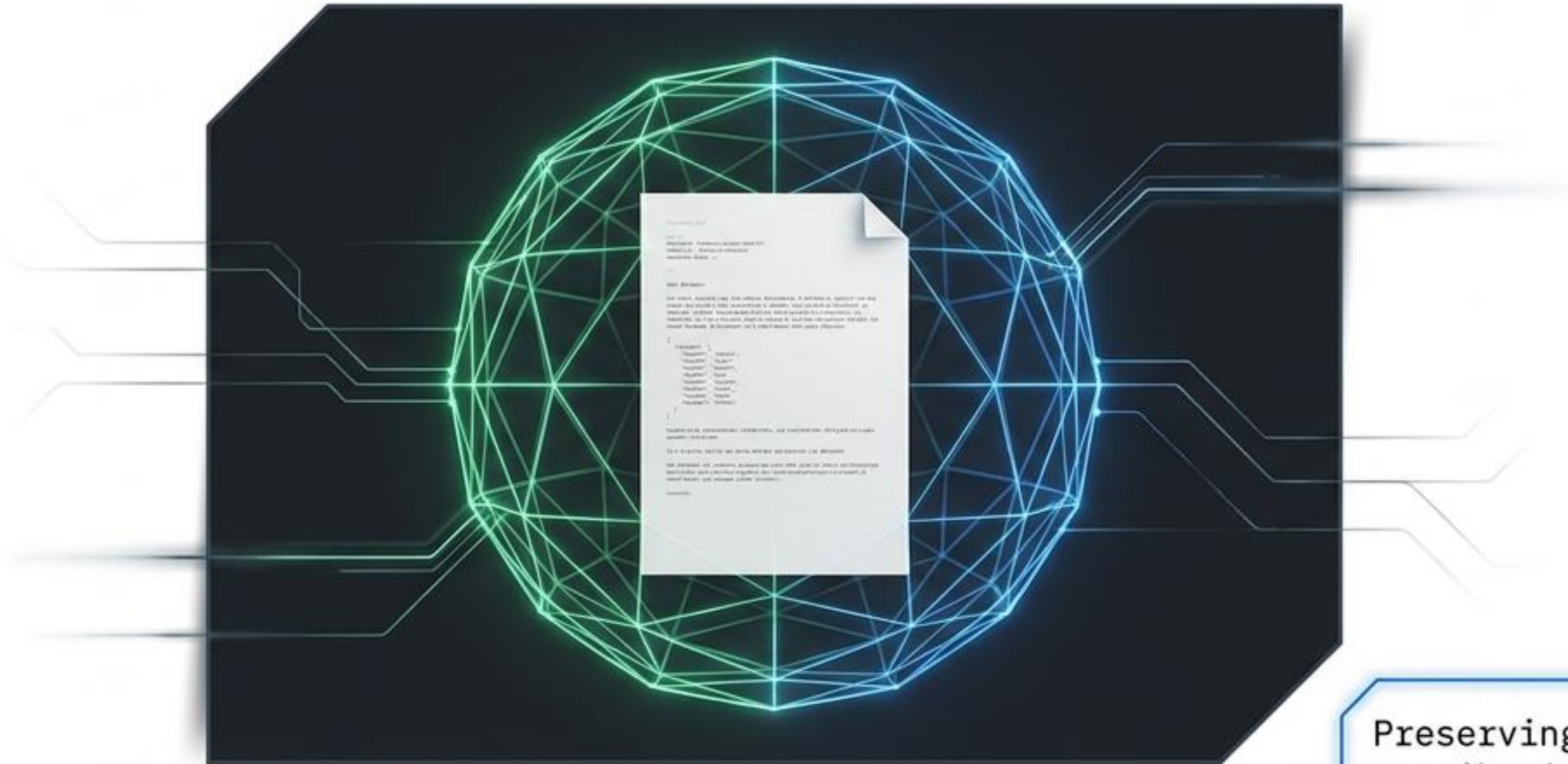


KNOBE Protocol v1

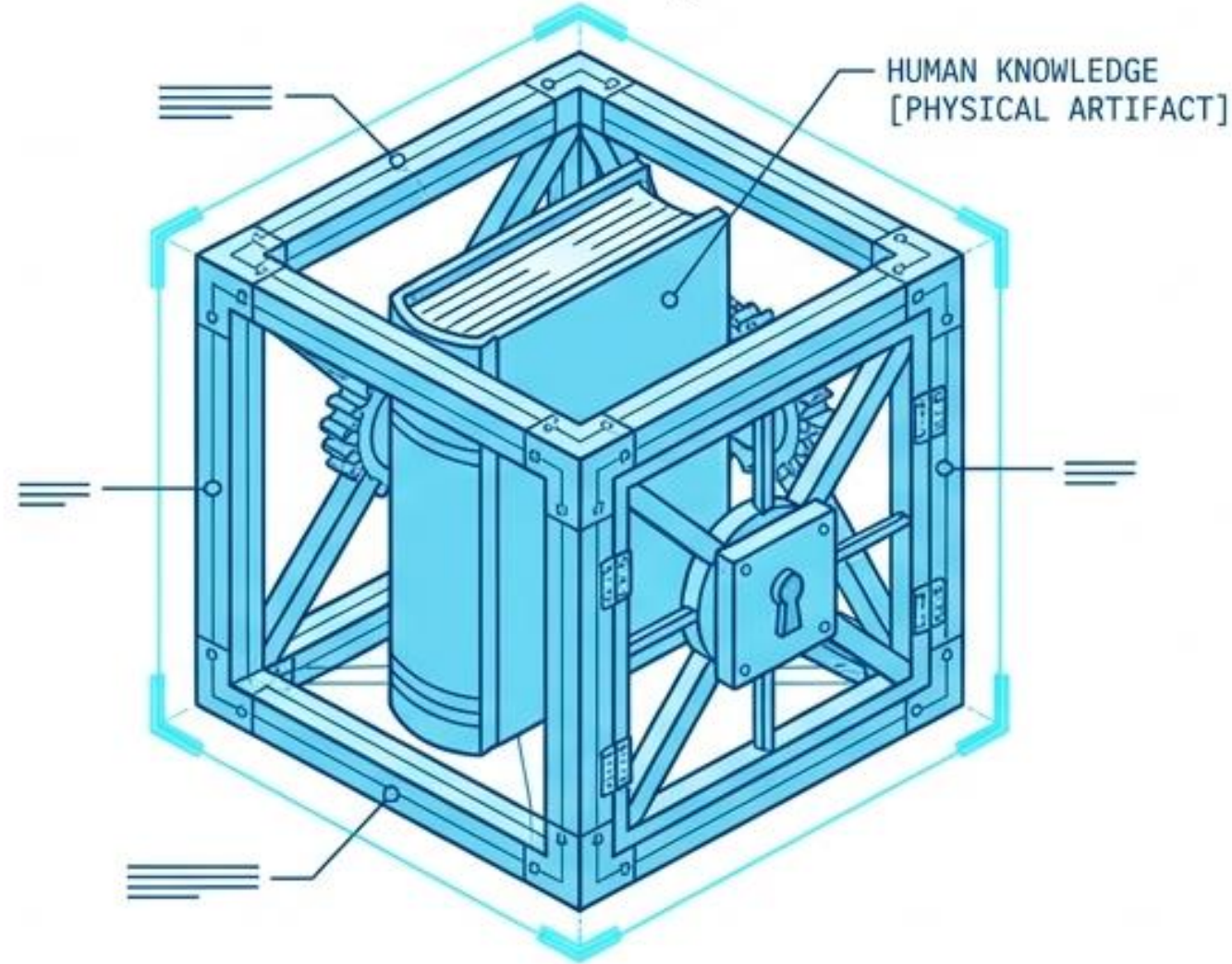
A Plain-Text Harness for Agentic Knowledge Work.



Preserving objecthood,
attribution, and
interpretive obligation
under compression.

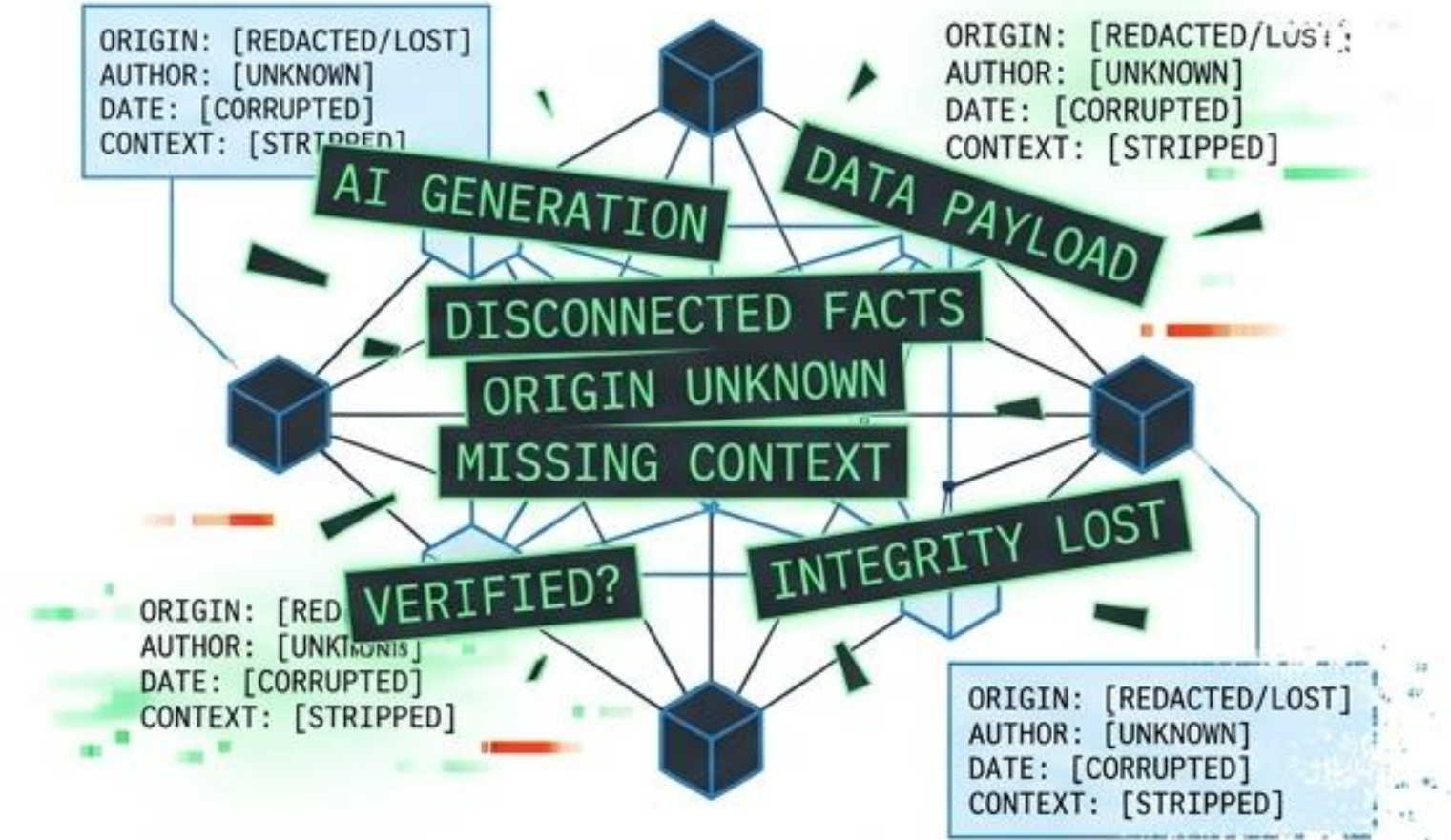
Institutions Solved the Wrong Problem for the AI Era

The Scarcity Era



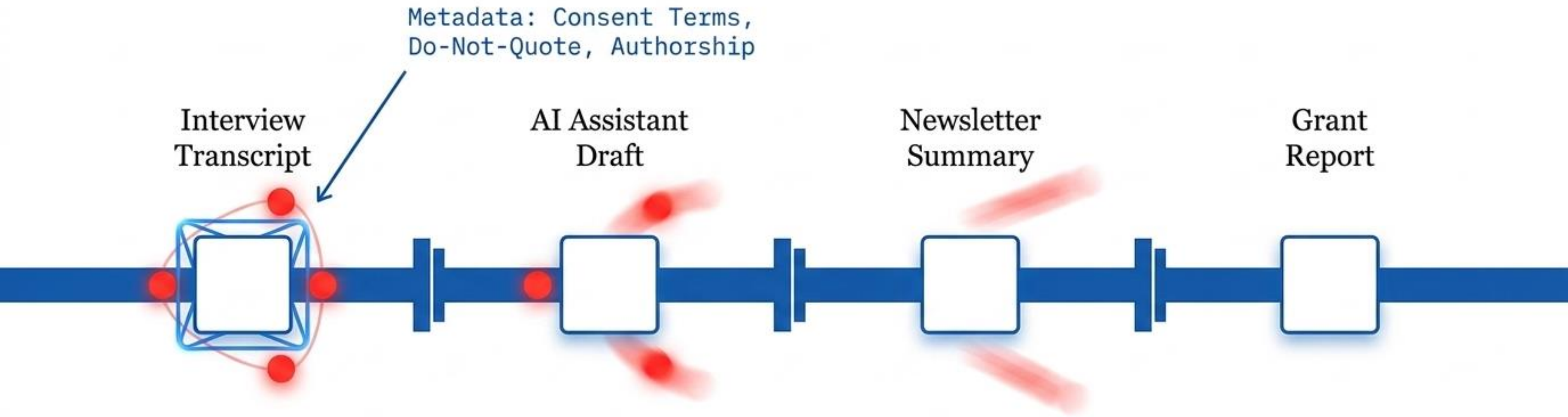
OLD PROBLEM: Scarcity
Too few copies, too little access.
Solution: Portability.

The Context Loss Era



NEW PROBLEM: Context Loss
Abundant copies, stripped of the conditions
that made them interpretable.

Context Survivorship Bias



The Illusion:

The fragment that survives transit is mistaken for the full object.

The Output:

Orphaned Fragments traveling without their interpretive obligations.

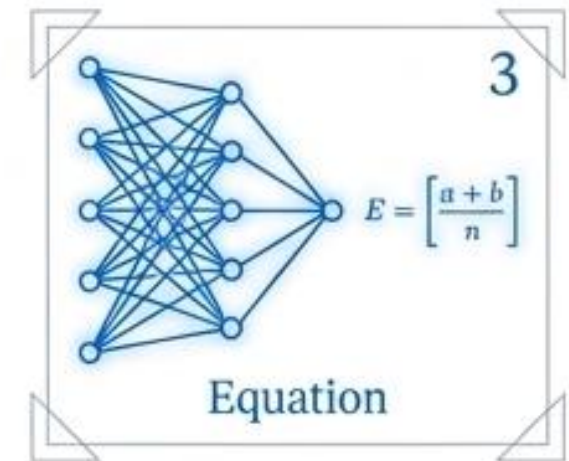
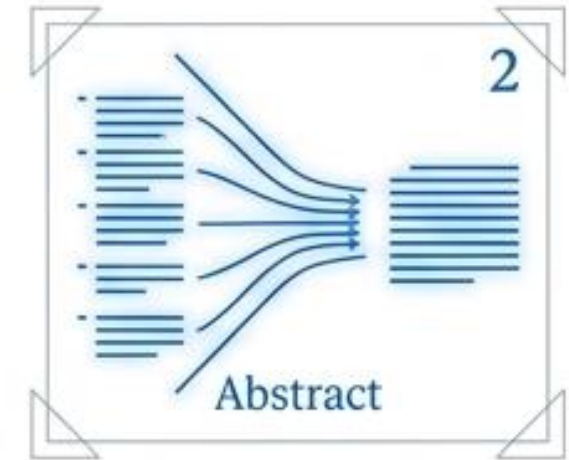
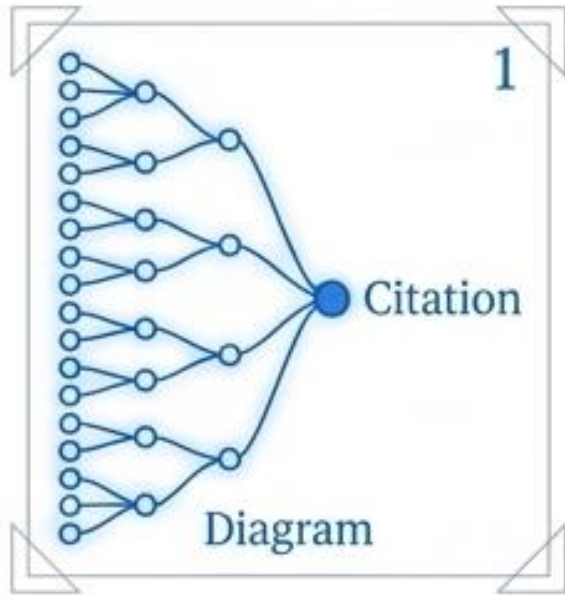
The Danger:

Zombie Knowledge circulating with unwarranted authority because its origins are no longer inspectable.

The False Diagnosis

~~The problem is
compression.~~

Compression without portable
interpretive obligation.



The Reality

1

Compression is unavoidable.
Citations, abstracts, and models
all compress.

The Real Problem

2

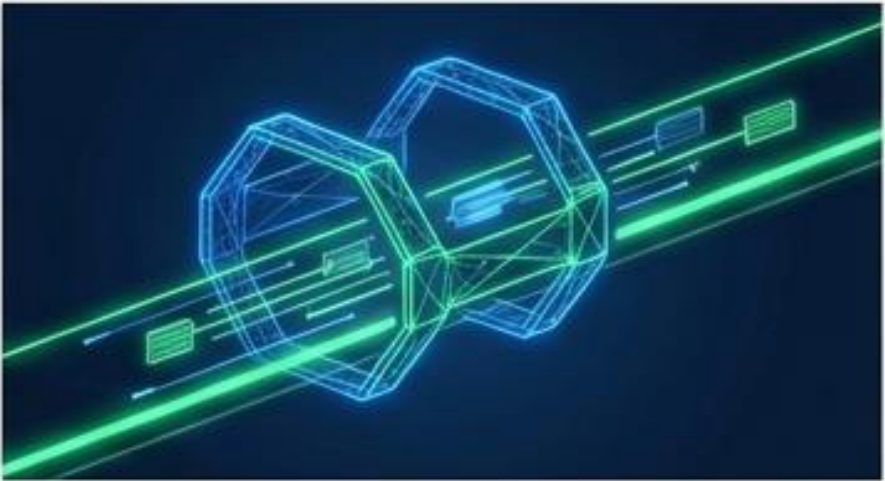
When context, provenance, and
fidelity limits are **severed** during
transit.

The Goal

3

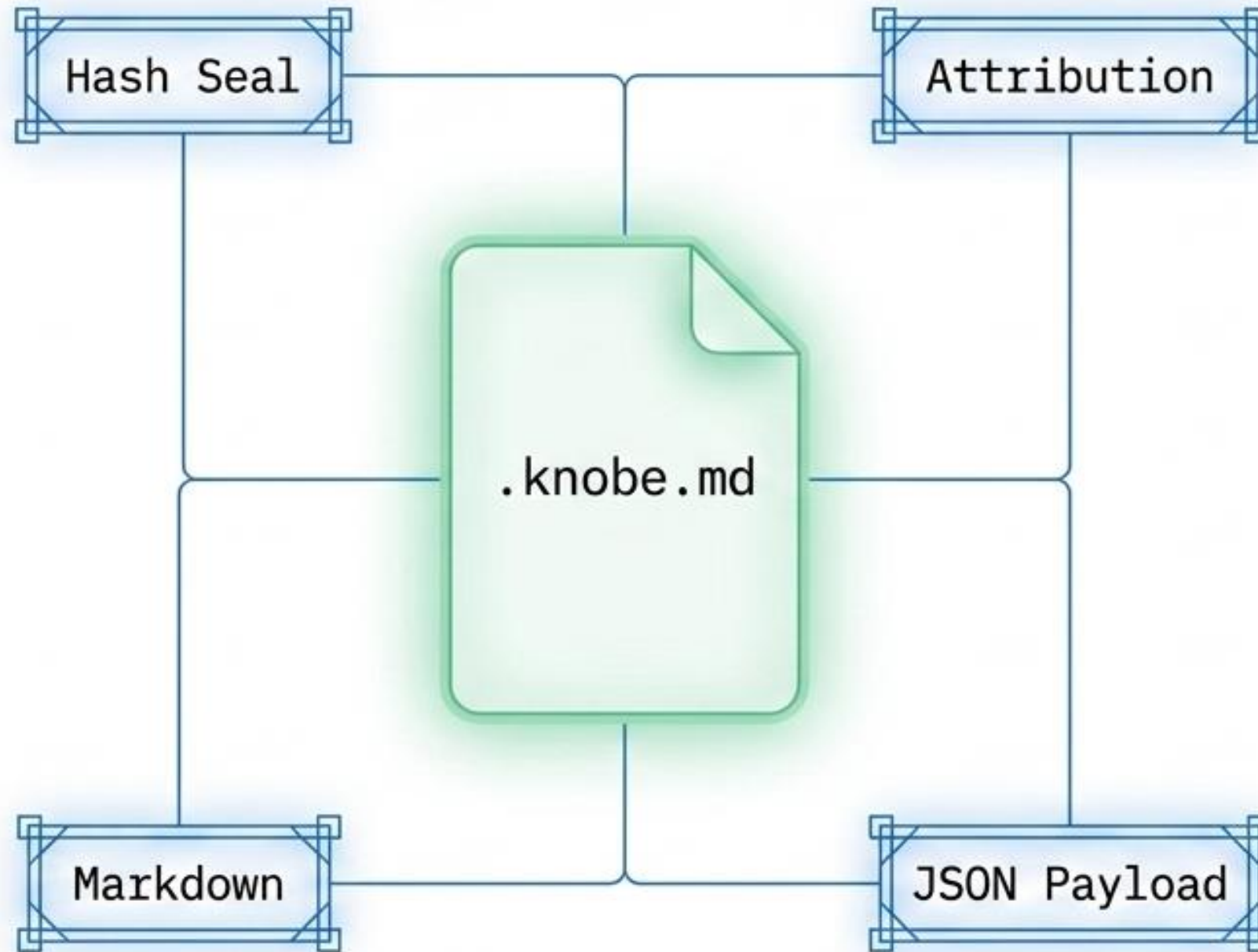
We do not need to stop compression.
We need artifacts that can still **answer**
for themselves after being **compressed**.

Systems of Record vs. Systems of Context

	Systems of Record 	Systems of Context (KNOBE) 
Function	Acts as Walls. Stores formal deposits and institutional approvals.	Acts as a Harness . Carries provenance and rules between systems.
Gravity	Objects must go TO them.	What the object WEARS as it moves.
Integration	Highly siloed by institutional boundaries.	Frictionless plain-text transit .

Insight: KNOBE replaces no system of record. It lowers coordination costs across all of them.

Introducing KNOBE Protocol v1



The Core Concept

An open, plain-text protocol. A single `.knope.md` file carries a human-readable document together with a machine-legible, hash-sealed record.

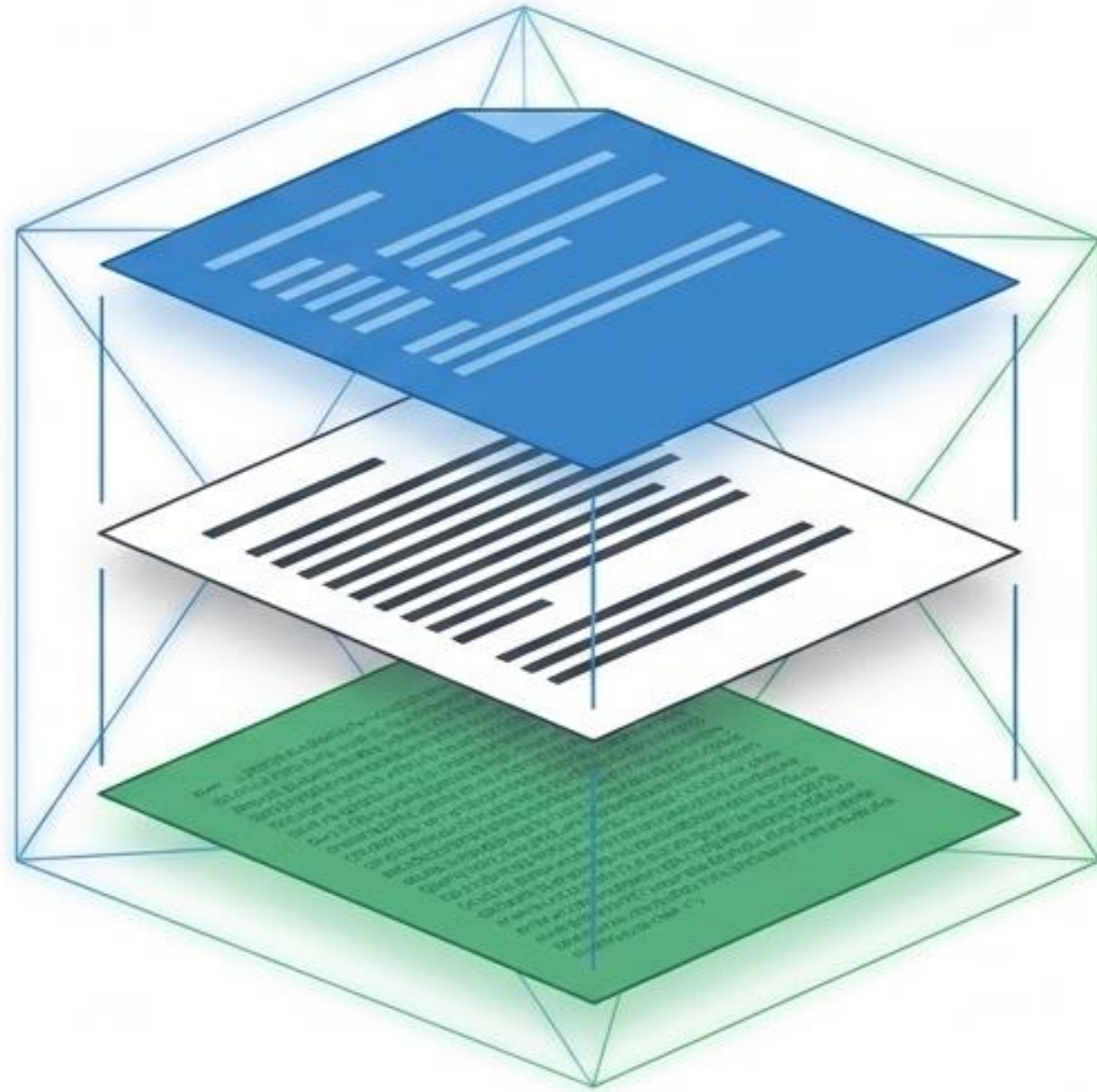
Dual Readability

Human-readable text + machine-verifiable payload coexist.

Zero Lock-In

Requires no proprietary platforms, no sidecar files, and no format conversions.

Anatomy of a KNOBE Artifact



Frontmatter (YAML)

Lightweight, human-readable metadata (Title, Author, License).

Body (Markdown)

Unconstrained text, easily readable in any standard editor or AI tool.

Payload (Base64 JSON)

The structured, machine-legible record. Hidden from casual reading, newline-anchored, and easily parsed.

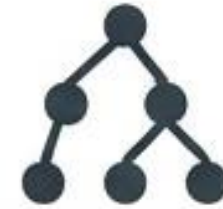
Takeaway: The object degrades gracefully. Handed to an agent, it is standard markdown. Handed to a parser, it is a verifiable cryptographic object.

The Four Rules of the Schema



1. Attribution Travels

`attribution.sources` names human and AI contributors, their roles, and rights-bearing status.



2. Lineage is Receipted

`parents[]` and `transformation_history[]` record derivations, forks, and adaptations by hash.



3. Integrity is Canonical

Cross-language SHA-256 tamper-evidence ensures payload bytes are identical to when sealed.



4. Interpretation Travels

`fidelity_limits` and `use_conditions` inform the receiving agent exactly how to treat the object.

Integrity $\neq$ Truth



What the hash proves

The payload bytes are identical to the moment the artifact was sealed.

What it does NOT prove

Truth, fairness, originality, or accuracy.

The Posture: Verification is cheap so human judgment can be spent where it is needed: evaluating claims.
The green check is where inspection **BEGINS**, not ends.

The Probabilistic Verification Fallacy

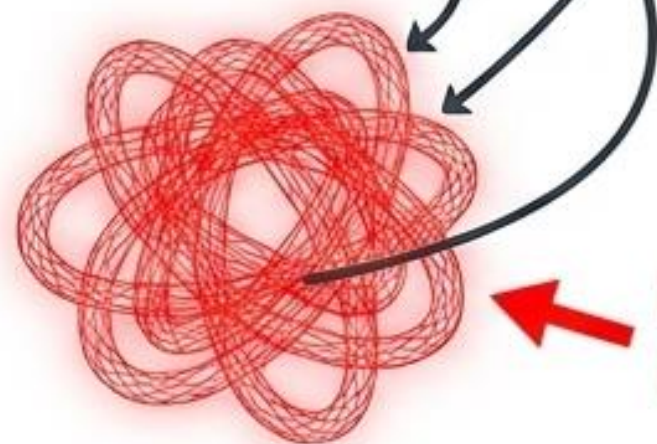
AI Verification Session

Verify hash: 85d886bb...

hash_check: 85d886bb...

verification: OK...

hash_check: 85d886bb...



Recursive
Blindness

SYSTEM_OUTPUT: NON_DETERMINISTIC_NARRATIVE

Step 1: The Fallacy



Asking an LLM to verify a hash results in “textual performance”—satisfying the narrative of verification—not cryptographic computation.

Step 2: Recursive Blindness



Models correctly diagnose their failure in text, then immediately hallucinate a fake hash in the next generation because no persistent state exists across the token gap.

Step 3: The Boundary

Deterministic Runtime



SHA-256
VERIFIED

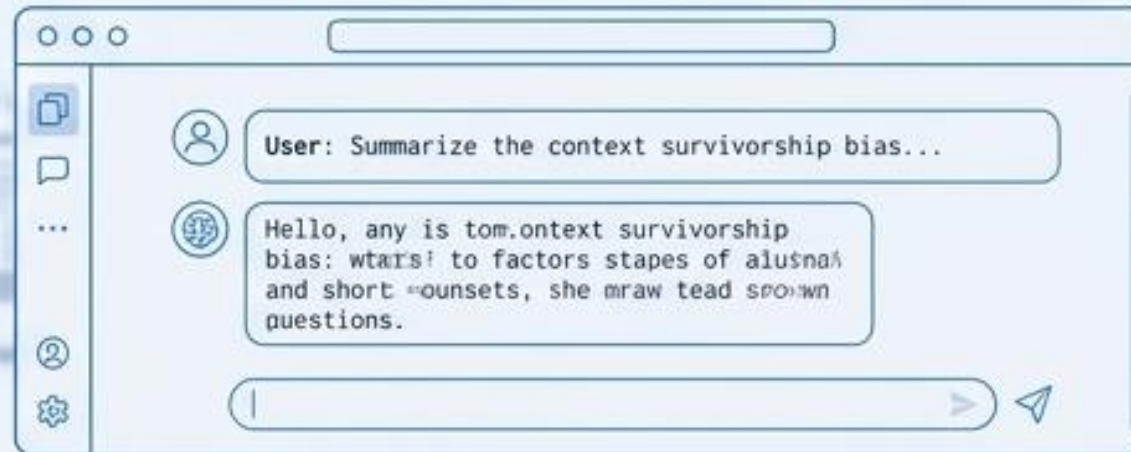
AI Probabilistic State



Cryptographic verification requires a deterministic runtime. This creates an architectural boundary that AI cannot fake.

Verification in Practice

NotebookLM (Synthesis Layer)



Can Answer:

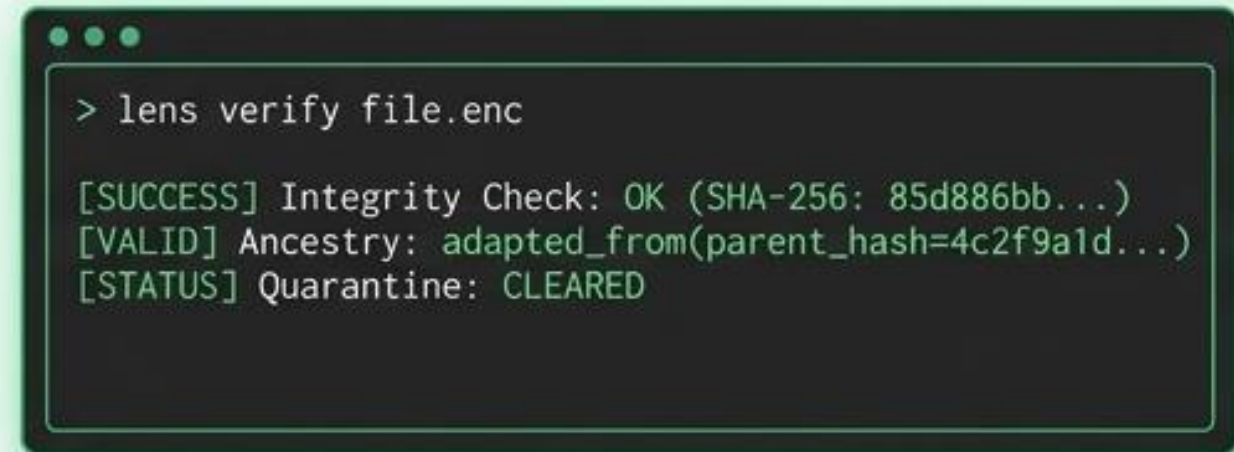
- ↳ What is context survivorship bias?
- ↳ What did the AI assist with?
- ↳ What is the summary?

Limitation:

Reads only the text layer.



Lens.py (Protocol Layer)



Can Prove:

- ✓ Has this file been altered?
- ✓ Is this adaptation faithfully linked to its parent hash (adapted_from)?
- ✓ What is the true quarantine_status?

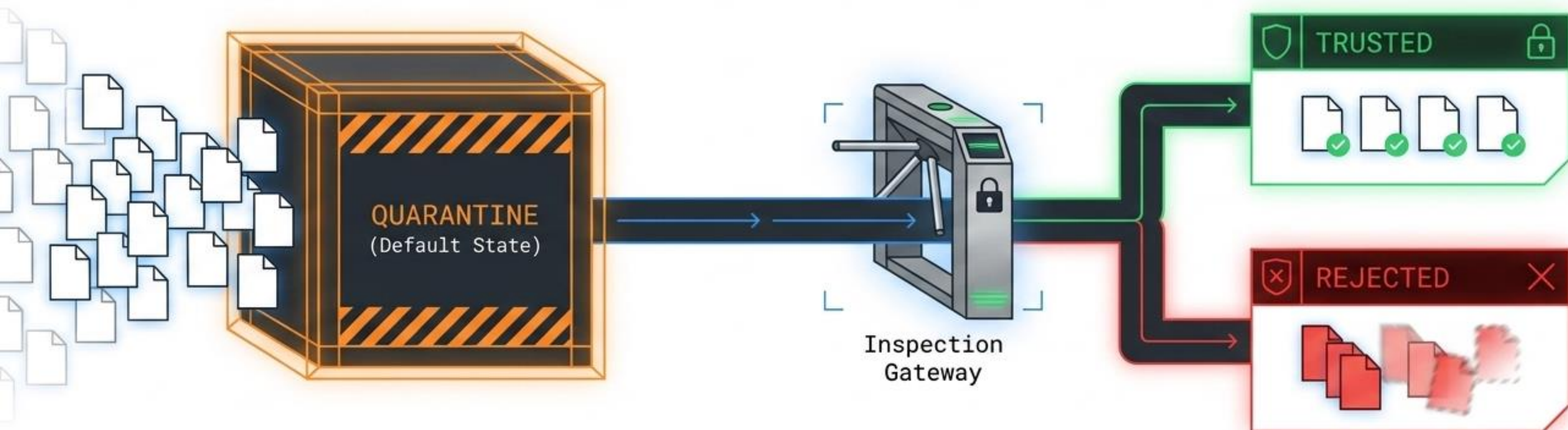
Strength:

Reads the deterministic state.



Conclusion: Synthesis models read the text; verifiers read the state. Both are necessary.

Trust Posture: Quarantine-First



Default State

Every new or external KNOBE defaults to `quarantine_status: quarantine`.

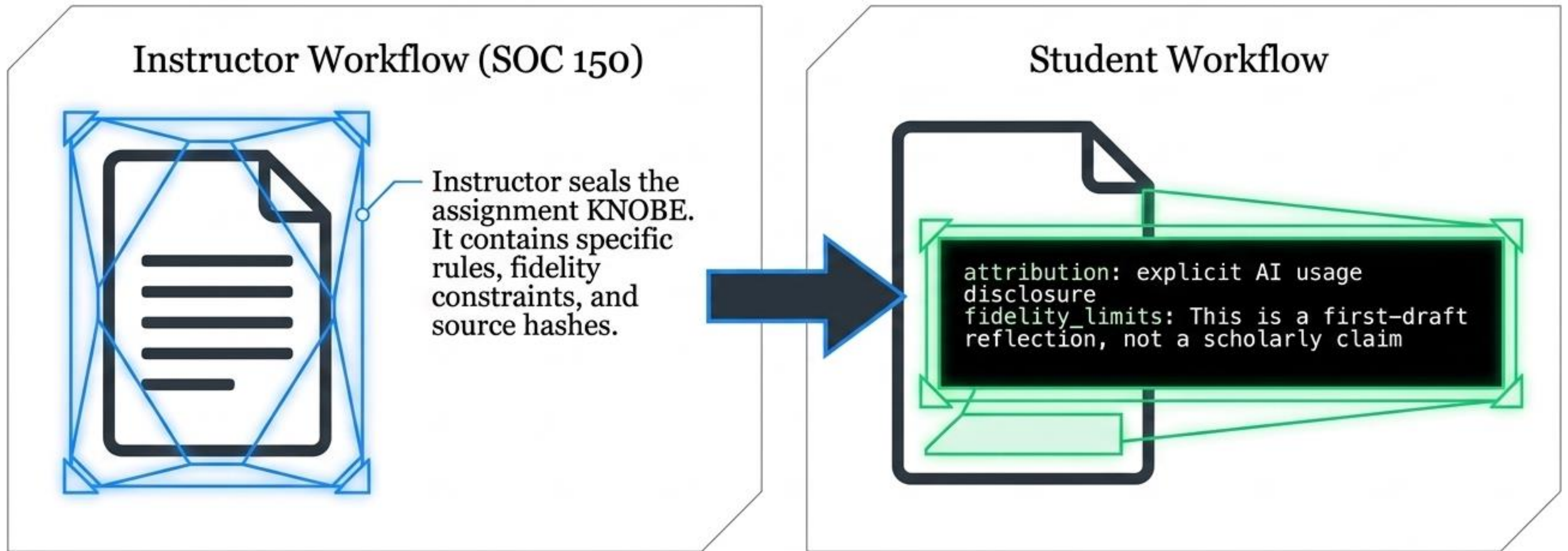
The Inversion

Typical systems treat arrival as implicit endorsement. KNOBE demands inspection before action.

Actionable Enforcement

Automated tools will not execute build recipes until explicitly marked trusted. Trust becomes a recorded decision.

Practical Application: Process Literacy



Execution: The student uses an AI for citation lookup but writes the text. Both human and AI contributions are explicitly categorized.

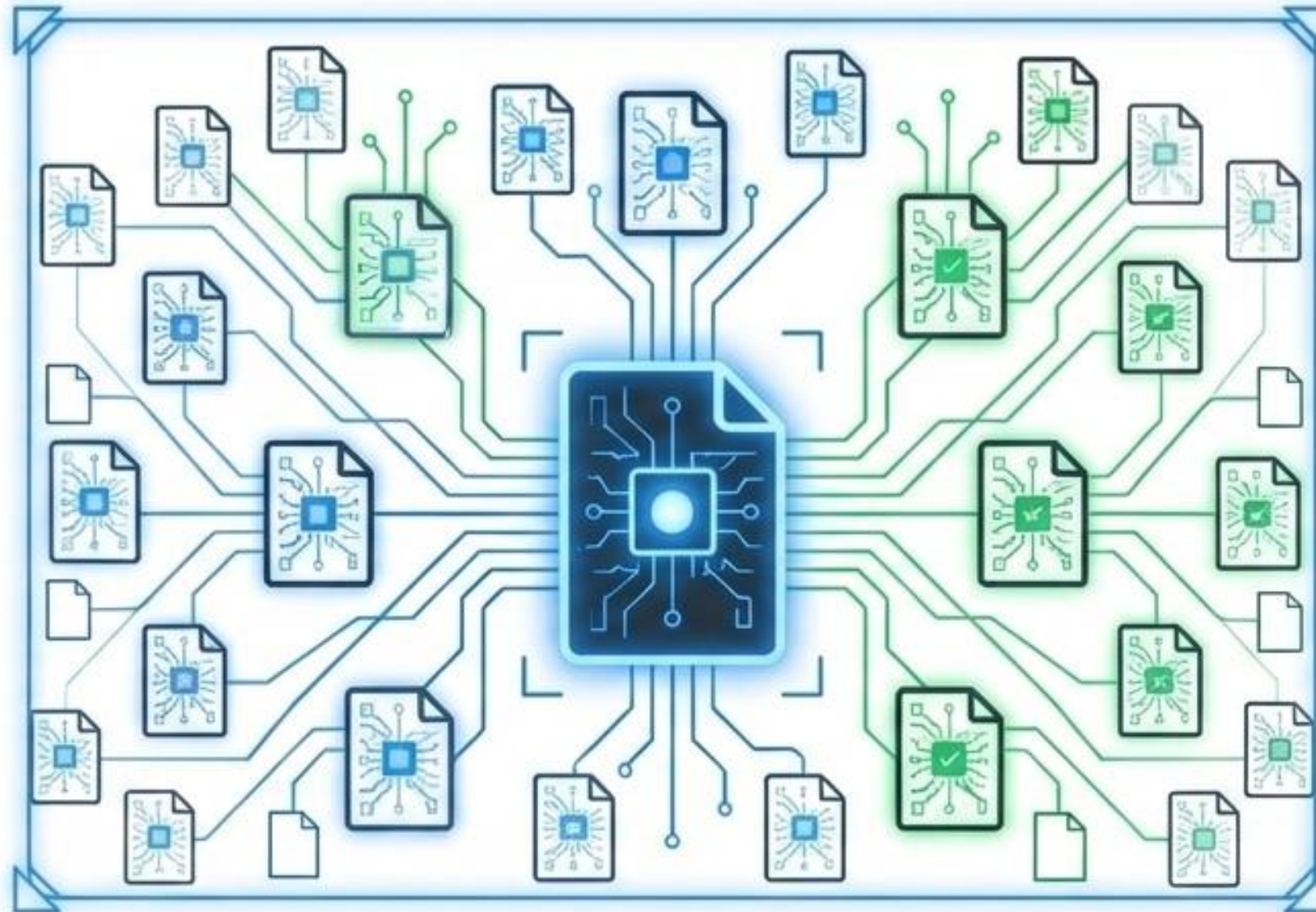
Value: Evaluates the pathway, not just the output. Allows decontextualization to be safely generative.

Fights the “Matthew Defect”: Prevents credit drift by permanently attaching attribution to the artifact.

Bootstrapping: Seeds & Groves

What is a Seed?

A self-describing KNOBE (`is_seed: true`) that teaches the protocol and carries build recipes to bootstrap local tools (viewers, (viewers, verifiers, combiners).

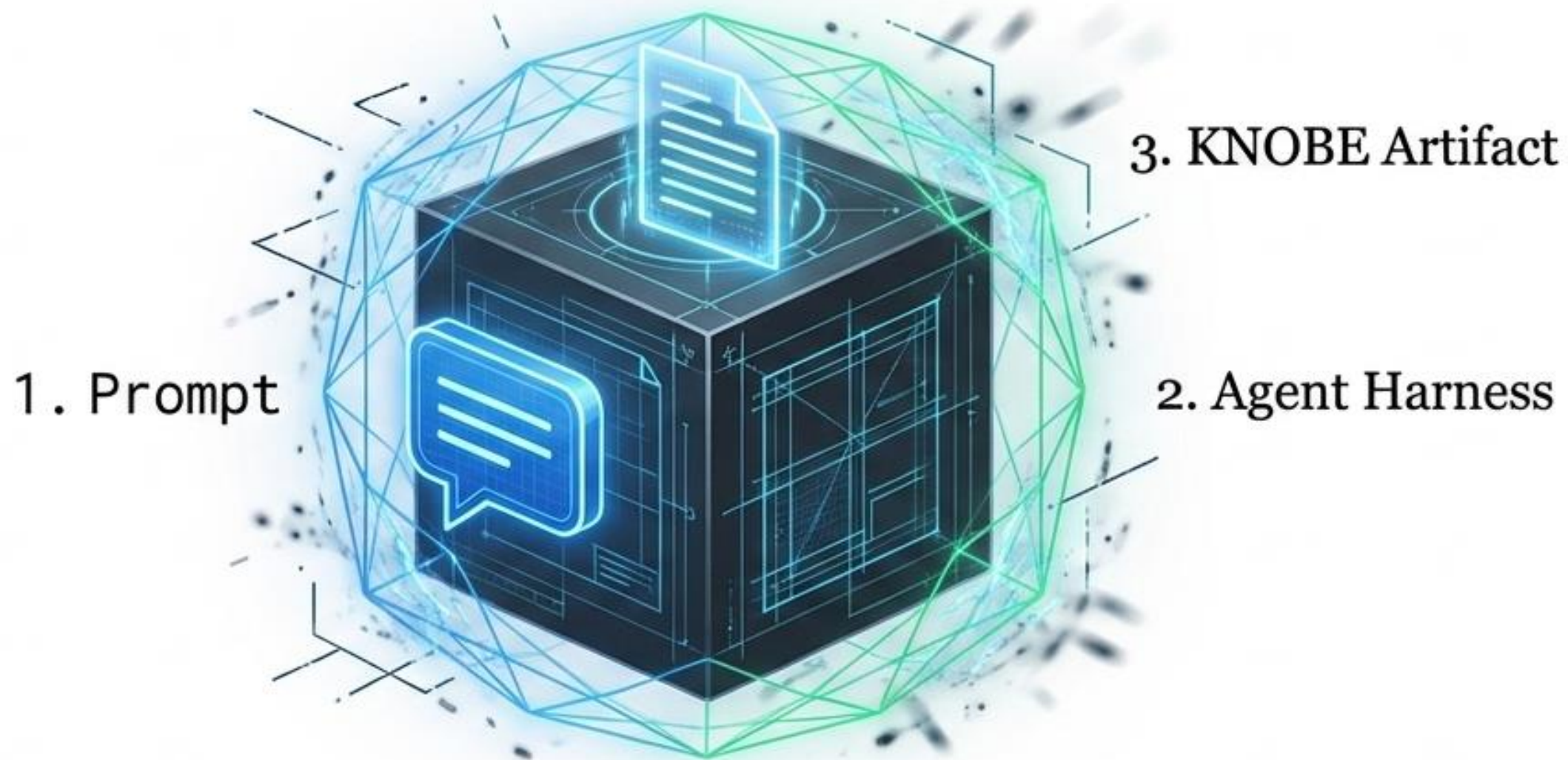


The Recursive Calibration Rule

Upward meta-levels must reduce rhetorical temperature and explicitly classify claims (verified, declared, inferred, speculative).

The Defense against “Recursive Contamination”
Prevents the failure mode where weak claims are dignified by abstraction until garbage looks load-bearing.

The Frontier: Agent Harness Engineering



Prompt Engineering	Harness Engineering	Agent Harness Engineering (KNOBE)
Governs what you ASK the agent.	Governs the agent's EXECUTION LAYER and policies.	Governs the OBJECT. The harness controls the encounter; the KNOBE preserves the object across encounters.